WORKSHOP SAFETY MANUAL

Apprentice and Entry Level Technician Orientation

MENOCLONE MOTORSPORTS | TRAINING EDITION | NOVEMBER 2026

A practical safety reference for apprentices learning vehicle maintenance, electrical diagnosis, and workshop procedures. Every task must be authorized, supervised as appropriate, and performed with the correct equipment.

Safety comes before the repair

No vehicle, deadline, or training exercise is worth an injury. Apprentices have the authority and responsibility to stop work when a condition appears unsafe. Tell the supervising technician, secure the area, and do not resume until the hazard has been addressed.

This manual is a starting point

This document supports shop orientation; it does not replace hands-on instruction, equipment manuals, vehicle service information, applicable regulations, or a shop-specific safety program. The supervising shop must assess its own hazards, provide required training and protective equipment, and establish emergency procedures.

The MENOCLONE safety promise

- I will ask before attempting work beyond my training or authorization.
- I will use the correct tools, protective equipment, and service information.
- I will report injuries, near misses, damaged equipment, and unsafe conditions promptly.
- I will protect coworkers, customers, vehicles, and the environment.

01 Orientation and shop rules

Before beginning any shift

• Check in with the assigned supervisor and confirm the tasks you may perform.

• Identify emergency exits, first aid supplies, eyewash facilities, spill kits, fire extinguishers, and the emergency meeting point.

• Learn where safety data sheets (SDS), equipment instructions, and vehicle service procedures are kept.

• Inspect your work area for slips, cords, leaks, poor lighting, and obstructed walkways.

• Wear fitted work clothing, secure long hair, remove loose jewelry, and wear closed-toe slip-resistant footwear.

Personal protective equipment

Hazard	Typical protection / action
Flying debris and splashes	Appropriate safety glasses; face shield when the task requires it.
Noise	Hearing protection when noise exposure requires it.
Sharp edges and handling	Task-appropriate gloves; avoid gloves near rotating machinery.
Chemicals	Chemical-compatible gloves and eye protection selected from the SDS.
Dust and fumes	Use required ventilation and respiratory protection only under a compliant program.

Conduct

Never work while impaired, distracted, or excessively fatigued. No horseplay. Keep food and drinks away from chemicals and workstations. Never bypass guards or safety interlocks. Follow the shop's rules on phones, smoking, and visitors.

02 Vehicle movement, lifts, and jacks

Vehicle movement

• Only authorized personnel may drive or reposition customer vehicles.

• Check the travel path, people nearby, mirrors, and blind spots before moving.

• Use a spotter when visibility or clearance is limited; agree on hand signals first.

• Secure the vehicle against unintended movement with the correct parking procedure and wheel chocks when required.

Raising a vehicle

• Use a lift or jack only after equipment-specific training and authorization.

• Confirm rated capacity, lift condition, correct vehicle lift points, and balanced positioning.

• Keep people clear during raising and lowering; check stability at a low height before continuing.

• Engage the lift's mechanical safety locks as instructed.

• Never work beneath a vehicle supported only by a hydraulic floor jack. Use properly rated jack stands at approved support points.

STOP WORK: If the vehicle shifts, lift arms move, a support point deforms, or any unusual sound occurs, clear the area and alert the supervisor.

Tires and wheels

Wheel and tire assemblies can store dangerous energy. Do not service split rims or inflate damaged tires without specialized training and approved equipment. Use correct wheel torque specifications and tightening sequences from vehicle service information.

03 Hand tools, power tools, and housekeeping

Before using a tool

• Choose the correct tool for the fastener and task; do not improvise extensions or use damaged tools.

• Inspect cords, plugs, guards, discs, bits, sockets, and air hoses.

• Keep hands away from pinch points and moving parts; clamp work securely.

• Disconnect power or air before changing accessories or clearing jams.

• Use eye and hearing protection appropriate to the task.

Rotating and cutting equipment

Grinding, drilling, and cutting can throw debris or catch clothing. Never remove a guard. Do not wear loose gloves or clothing near rotating parts. Confirm accessories are rated for the tool and use the manufacturer's operating procedure.

Shop housekeeping

Clean spills promptly using approved methods. Return tools after use, route cords away from walkways, store heavy items low, and keep emergency exits clear. Dispose of used rags and contaminated materials in designated containers.

Compressed air

Never direct compressed air at skin or clothing. Use approved nozzles and pressure limits. Do not use air to blow brake dust or other potentially hazardous dust into the shop.

04 Chemicals, fluids, and fire prevention

Know the material before opening it

• Read the product label and SDS; identify the main hazards and required PPE.

• Keep chemicals in labeled, approved containers; never mix unknown products.

• Use ventilation appropriate to cleaners, solvents, fuels, and adhesives.

• Keep ignition sources away from flammable liquids and vapors.

• Wash exposed skin and report spills or exposures immediately.

Automotive fluids

Material	Key precaution
Gasoline and solvents	Flammable vapors; eliminate ignition sources and control spills.
Coolant	May be toxic; prevent ingestion and clean spills promptly.
Brake fluid	Avoid skin and eye contact; protect painted surfaces.
Battery electrolyte	Corrosive; use appropriate eye and hand protection.
Used oil	Avoid repeated skin contact; collect and recycle properly.

Fire response

Know the shop's emergency plan. Raise the alarm, evacuate when necessary, and call emergency services. Only trained personnel should use an extinguisher, only on a small, contained fire with a safe exit behind them. Never fight a spreading fire.

05 Electrical systems and batteries

12-volt vehicle electrical work

• Identify the circuit and consult the correct wiring diagram before testing.

• Switch off power and follow the vehicle's specified battery isolation procedure before work that requires it.

• Use properly rated test equipment; inspect meter leads and use the correct meter setting.

• Protect circuits with the specified fuse; never bypass protection with wire or foil.

• Avoid shorting battery terminals with metal tools, jewelry, or test leads.

Battery handling

Lead-acid batteries can produce explosive gas and contain corrosive electrolyte. Use eye protection, ventilation, and approved lifting methods. Follow the vehicle and charger manufacturer's connection procedure. Never jump-start a visibly damaged, leaking, or frozen battery.

Hybrid and electric vehicles

HIGH VOLTAGE — AUTHORIZED PERSONNEL ONLY. Orange cables and high-voltage components may remain dangerous even after the vehicle is switched off. Apprentices must not disconnect, probe, or service high-voltage components without the required vehicle-specific training, equipment, isolation procedure, and authorization.

Airbags and restraint systems

Supplemental restraint systems can deploy unexpectedly. Do not probe or disconnect airbag or pretensioner circuits unless trained and following the exact service procedure.

06 Engine, exhaust, brakes, and moving parts

Engines and exhaust

• Keep hands, hair, and clothing away from belts, fans, and rotating parts; electric fans may start unexpectedly.

• Allow hot engines, exhaust parts, coolant, and oil to cool before contact.

• Never open a hot pressurized cooling system.

• Do not run engines indoors without a properly functioning exhaust extraction system.

Brake systems

Brake dust may contain hazardous materials. Do not dry-brush or blow dust with compressed air. Follow approved wet-cleaning or HEPA-filtered collection procedures and the shop's respiratory protection requirements. Use correct support methods and service specifications.

Stored energy

Suspension springs, pressurized fuel systems, hydraulic systems, and refrigerant circuits can release energy suddenly. Apprentices must not open or disassemble these systems without specific instruction and authorization. Air-conditioning refrigerant service requires applicable certification and approved recovery equipment.

Road tests

Only authorized, licensed drivers may road-test vehicles. Confirm insurance and shop procedures, use seat belts, obey traffic laws, and never perform distracting diagnostic actions while driving.

07 Diagnostics, OBD-II, and electronics benches

Safe diagnostic habits

• Confirm vehicle identification and the appropriate service procedure before connecting equipment.

• Inspect OBD-II connectors, adapters, cables, and scan tools for damage.

• Use only equipment intended for the vehicle's electrical system.

• Keep cables clear of pedals, steering controls, hot parts, and moving components.

• Do not clear diagnostic trouble codes, reset modules, or alter settings without authorization.

Bench electronics and prototype work

MENOCLONE Motorsports projects such as the MNC UNO R1 and MNC DIAG-01 are learning and development platforms. Prototype electronics must be tested on an isolated, current-limited bench supply where appropriate, with verified voltage and polarity. Never connect unvalidated boards to a customer vehicle.

Diagnostic responsibility

A fault code identifies a system or condition that requires investigation; it does not prove a particular component is defective. Record findings accurately and consult a supervising technician before recommending repairs.

08 Incident response and stop-work rules

Stop work immediately when

• A vehicle, lift, jack stand, or hoist appears unstable.

• You smell strong fuel vapors, see smoke, or suspect a fire.

• A person is injured, exposed to a chemical, or becomes ill.

• Electrical equipment sparks, overheats, or behaves unexpectedly.

• You encounter a task beyond your training or do not understand the procedure.

If an incident occurs

Protect yourself first. Alert others and the supervisor, call emergency services when needed, and follow the site's emergency plan. Do not move an injured person unless they are in immediate danger. Only provide first aid within your training. Preserve the scene when safe so the incident can be reviewed.

Near misses matter

Report events that almost caused injury or damage. Record what happened, the conditions present, and actions taken. A near miss is an opportunity to improve procedures before someone is hurt.

Emergency contact information — complete at orientation

Shop / location	_____
Supervisor	_____
Emergency phone	_____
Meeting point	_____
First aid / eyewash	_____

09 Apprentice safety checklist

The supervising technician should review each item with the apprentice. Mark completion only after the apprentice can demonstrate the required safe behavior.

☐ Locates exits, first aid, eyewash, spill kit, and fire alarm.

☐ Identifies correct PPE and reads chemical labels / SDS.

☐ Demonstrates safe hand tool inspection and workstation setup.

☐ Explains vehicle movement rules and wheel chock use.

☐ Identifies lift points and explains jack stand requirements.

☐ Explains battery hazards and safe 12-volt testing.

☐ Recognizes high-voltage and airbag restricted work.

☐ Explains engine heat, moving parts, and exhaust risks.

☐ Explains brake dust controls and chemical spill reporting.

☐ Explains OBD-II tool checks and prototype bench restrictions.

☐ Can describe stop-work authority and incident reporting.

Supervisor assessment

Areas requiring additional training: _____

_____

Practical demonstration completed on: _____

Supervisor / evaluator: _____

10 Safety acknowledgment and sign-off

I have reviewed the MENOCLONE Motorsports Apprentice Workshop Safety Manual and discussed its requirements with the supervising evaluator. I understand that I must follow shop-specific rules, ask for guidance when unsure, use required protective equipment, and stop work when conditions are unsafe.

I understand that this acknowledgment records orientation only. It does not certify competence, authorize unsupervised work, establish employment, or replace task-specific training.

Apprentice full name: _____

Apprentice signature: _____

Date: _____

Certificate evaluator / supervising technician:

_____

Evaluator signature: _____

Date: _____

Shop / training location: _____

Evaluator confirmation

I confirm that the apprentice has been oriented to the topics identified in this manual and that any additional task-specific training and supervision requirements have been explained.

Training status: ☐ Orientation complete ☐ Additional instruction required

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This manual is not a government-issued safety certification or a substitute for workplace compliance requirements.